MILITARY SPECIFICATION

JOINT BAR, RAIL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 24-inch (610-mm) long, 4 hole, steel joint bars used in pairs to join rail ends.

1.2 Classification. Joint bars shall be for the following weight-rails, as specified:

60-pound (27-kg) ASCE rail.
70-pound (32-kg) ASCE rail.
90-pound (41-kg) ARA-A rail.
115-pound (52-kg) AREA rail.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 2250

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:44Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

QQ-W-461 - Wire, Steel, Carbon (Round, Bare, and Coated).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking For Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN RAILWAY ENGINEERING ASSOCIATION (AREA)

Specifications for High-Carbon or Quenched-Carbon Steel Joint Bars.

(Application for copies should be addressed to the American Railway Engineering Association, 59 East Van Buren Street, Chicago, IL 60605.)

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

Specifications for High-Carbon or Quenched-Carbon Steel Joint Bars.

(Application for copies should be addressed to the American Society of Civil Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A3 - Low-Carbon Steel Joint Bars.
A4 - Medium-Carbon Steel Joint Bars.
A5 - High-Carbon Steel Joint Bars.
A49 - Quenched-Carbon Steel Joint Bars.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:44Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department 1616 P Street NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago IL 60606.)

3. REQUIREMENTS

3.1 Description. The joint bars shall be fabricated in accordance with American Railway Engineering Association (AREA) specifications, American Society of Civil Engineers (ASCE) specifications, and ASTM A3, A4, A5, and A49 as applicable for steel joint bars. Joint bars shall be head-free or head-bearing and long-toe or short-toe, as specified.

3.2 Sections. Joint bar sections and punching shall be in accordance with Figure 1. Dimensional tolerances shall be in accordance with ASTM A3, A4, A5, or A49 as applicable.

3.3 Identification marking. The joint bars shall be identified in accordance with the contractor's standard practice.

3.4 Materials. Materials shall conform to applicable AAR and commercial specifications currently used by the railroad industry for this type of equipment unless otherwise specified herein (see 6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:44Z
Check the source to verify that this is the current version before use.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.4).

4.3 Quality conformance inspection.

4.3.1 Examinations and tests. Joint bars shall be examined and tested in accordance with AREA, ASCE, and ASTM specifications as applicable for steel joint bars. When specified, a check analysis shall be performed. Nonconformance to 3.1 or 3.2 shall be cause for rejection.

4.4 Inspection of packaging.

4.4.1 Quality conformance inspection of pack.

4.4.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.4.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.1.3 Examination. Samples selected in accordance with 4.4.1.2 shall be examined for the characteristics listed in Table I. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

Source: https://assist.dla.mil -- Downloaded: 20261O-17T22:44Z.
Check the source to verify that this is the current version before use.

TABLE I
EXAMINATION SCHEDULE

NUMBER	CHARACTERISTICS	REQUIREMENT PARAGRAPH
Major		
101	Packing not Level A or Commercial as specified. (see 6.2)	5.1
102	Joint bars not packed for Level A as specified.	5.1.1
103	Joint bars not packed for commercial as specified.	5.1.2
Minor		
201	Marking for military level of packaging not in accordance with MIL-STD-129.	5.2.1
202	Marking for commercial level of packaging not as specified.	5.2.2

5. PACKAGING

5.1 Packing. Packing shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A. Joint bars of like description shall be secured together, in quantities not to exceed 150 pounds (68 kg) gross weight, with soft annealed wire conforming to QQ-W-461, Finish 5, Class 2, composition, tensile strength, and wire size as applicable.

5.1.2 Commercial. Joint bars of like description shall be prepared for shipment in a manner to assure carrier acceptance and delivery to destination, without damage, at the lowest ratings of the carrier in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking.

5.2.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.2.2 Commercial packaging. Marking shall reflect as a minimum, the item name, National Stock Number, quantity, contract number, and appropriate

Source: https://assist.dla.mil -- Downloaded: 202610-17T22:44Z
Check the source to verify that this is the current version before use.

address markings. The marking shall be applied with a waterproof material, be legible, and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The joint bars are intended for use in laying of tracks to join rail ends in position with reference to surface and gage side alinement. Bar and rail should be used in the following combinations:

Joint bar for 60-pound- (27-kg) ASCE rail.
Joint bar for 70-pound- (32-kg) ASCE rail.
Joint bar for 90-pound- (41-kg) ARA-A rail.
Joint bar for 115-pound- (52-kg) AREA rail.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Joint bars required for specified weight-rail (see 1.2).
(c) Whether joint bars shall be head-free or head-bearing, long-toe, or short-toe (see 3.1).
(d) Whether a check analysis is required (see 4.3.1).
(e) Degree of packing required (see 5.1).

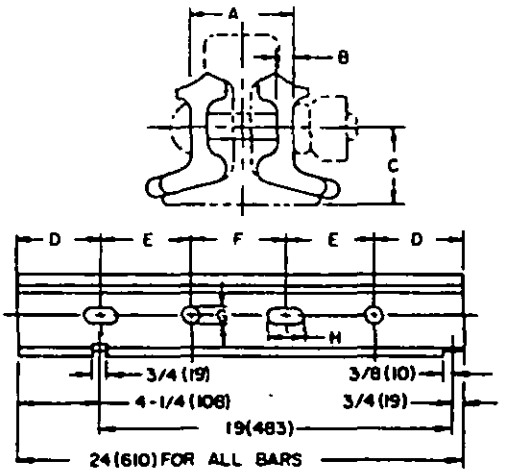
6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

Custodians:
Army - ME
Navy - YD

Preparing activity:
Army - ME

Project 2250-0051

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:44Z
Check the source to verify that this is the current version before use.



NOTE: DIMENSIONS ARE IN INCHES (MILLIMETERS).

JOINT BAR PUNCHING

ENGLISH SYSTEM, dimensions are in inches (in).

TO BE USED WITH	A	B (MIN)	C	D	E	F	G	H
60 LB. ASCE RAIL	2-13/16 ± 1/8	7/16	1-115/128	4-15/32	5	5-1/16	7/8	1-1/8
70 LB. ASCE RAIL	2-27/32 ± 1/8	7/16	2-3/64	4-15/32	5	5-1/16	7/8	1-1/8
90 LB. ARA-A RAIL	3-3/8 ± 1/8	7/16	2-37/64	3-3/4	5-1/2	5-1/2	1-1/16	1-13/32
115 LB. AREA RAIL	4 OR 3-5/8	7/16	2-7/8	2-7/16	6	7-1/8	1-1/16	1-13/32

METRIC SYSTEM, dimensions are in millimeters (mm).

TO BE USED WITH	A	B	C	D	E	F	G	H
27 kg ASCE RAIL	71 ± 3	11	48	114	127	129	22	29
32 kg ASCE RAIL	72 ± 3	11	52	114	127	129	22	29
41 kg ARA-A RAIL	86 ± 3	11	65	95	140	140	27	36
52 kg AREA RAIL	102 OR 92	11	73	62	152	181	27	36

FIGURE 1. Joint bar section.

X-2015B

Source: https://assist.dla.mil -- Downloaded: 202640-17T22:44Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-J-12368D Joint Bar, Rail

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:44Z
Check the source to verify that this is the current version before use.